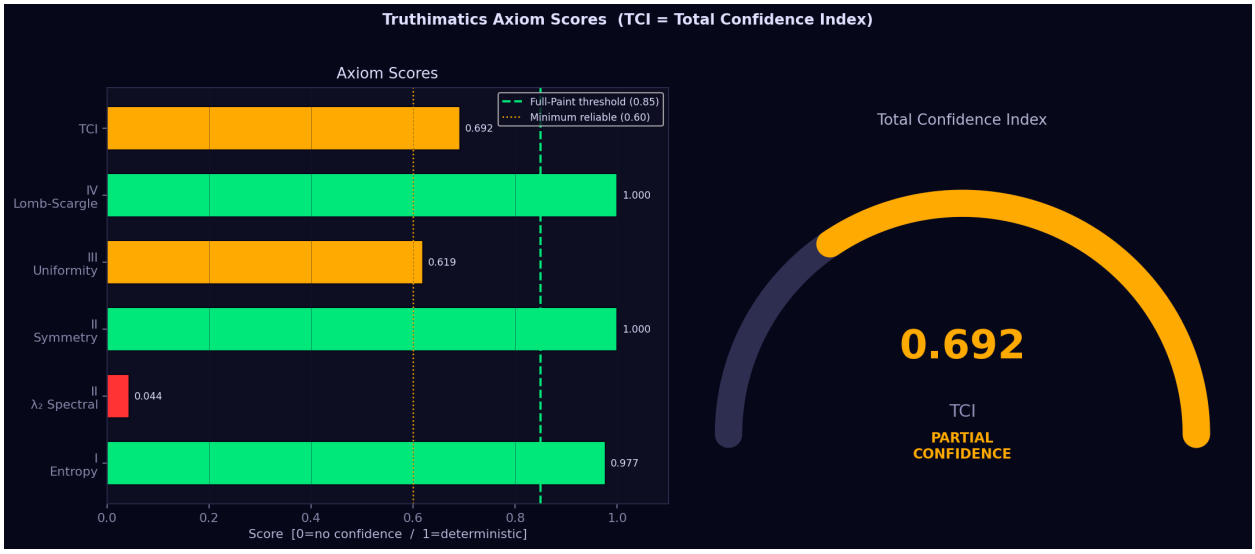
# AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

| | |
|---|---|
| **Specimen ID** | AXIOM_8C95B2BE3D |
| **Taxa** | Iguanodon |
| **Clade** | Ornithischian |
| **PBDB Records** | 133 |
| **Mass (regression)** | 3641.1 kg (Campione & Evans 2012) |
| **Mass (volumetric)** | 3800.0 kg (Silhouette integral, ρ=1060.0 kg/m³) |
| **Body Volume** | 4.000 m³ |
| **Validity Score** | 1.0000 / 1.0000 |
| **TCI** | 0.6916 (partial) |
| **Axiom I Entropy** | 0.9771 |
| **Axiom II λ■** | 0.1763 (symmetry 100%) |
| **Axiom III Uniformity** | 0.6187 |
| **Axiom IV Lomb FAP** | 1.000000 peak 0.000 Hz |
| **Integrity** | 56.0% Measured / 44.0% Inferred |
| **Elements** | 25 |
| **CoG (m)** | [0.277, 0.863, 0.0] |
| **Generated** | 2026-05-01 00:01 UTC |

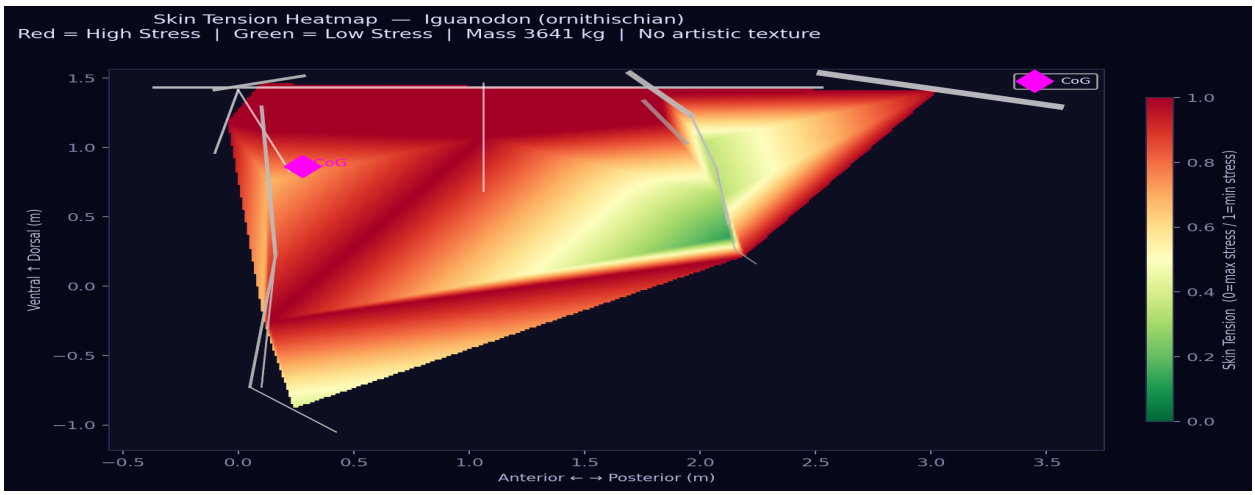
# 1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



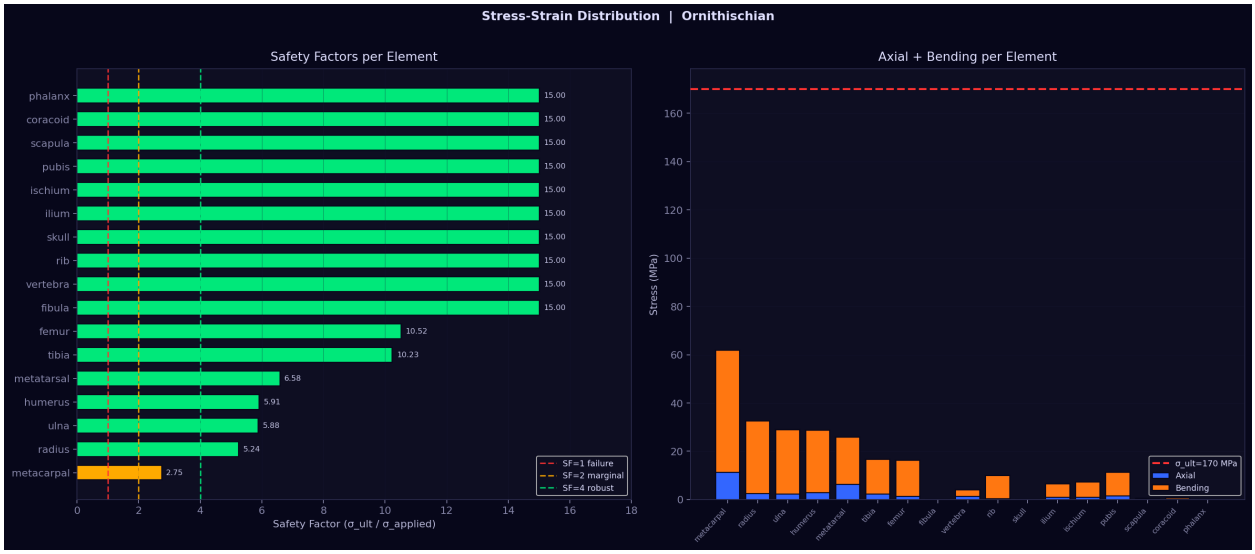
## 2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



# 3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

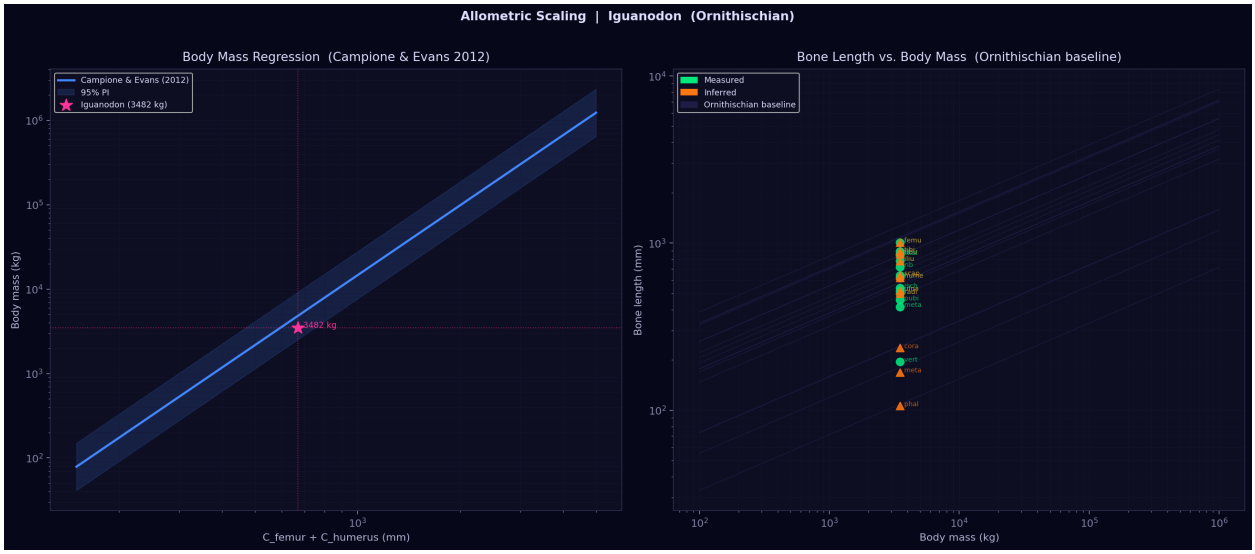



## Per-Element Stress Summary

| Element | σ_ax MPa | σ_bend MPa | σ_total MPa | SF | Status |
|---|---|---|---|---|---|
| coracoid | 0.251 | 0.460 | 0.710 | 15.000 | OK |
| femur | 1.364 | 14.799 | 16.163 | 10.518 | OK |
| fibula | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| humerus | 2.753 | 26.036 | 28.789 | 5.905 | OK |
| ilium | 0.973 | 5.509 | 6.482 | 15.000 | OK |
| ischium | 0.964 | 6.349 | 7.313 | 15.000 | OK |
| metacarpal | 11.365 | 50.498 | 61.863 | 2.748 | WARNING |
| metatarsal | 6.338 | 19.478 | 25.816 | 6.585 | OK |
| phalanx | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| pubis | 1.439 | 9.768 | 11.207 | 15.000 | OK |
| radius | 2.467 | 29.981 | 32.448 | 5.239 | OK |
| rib | 0.290 | 9.612 | 9.902 | 15.000 | OK |
| scapula | 0.135 | 0.494 | 0.629 | 15.000 | OK |
| skull | 0.000 | 0.000 | 0.000 | 15.000 | OK |
| tibia | 2.362 | 14.259 | 16.622 | 10.228 | OK |
| ulna | 2.218 | 26.694 | 28.913 | 5.880 | OK |
| vertebra | 1.301 | 2.661 | 3.962 | 15.000 | OK |

# 4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 ($R^2$=0.97; N=426 non-avian dinosaurs).



## 5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

| Parameter | Value | Notes |
|---|---|---|
| **Hip Height** | 2.315 m | Femur + tibia + metatarsal |
| **Estimated Speed** | 14.2 km/h  (3.96 m/s) | Gait: Walk \| Fr=0.689 |
| **Stride Length** | 4.75 m | Alexander (1976) / Thulborn (1990) |
| **Bite Force** | 14999.1 N  (1529.0 kgf) | Ornithischian regression (Norman & Weishampel 1991) |
| **Skull Length** | 850.0 mm | Mid-shaft proxy from skull feature vector |
| **Femoral Robusticity** | 0.3888 | Circumference / length (dimensionless) |
| **Limb Strength Index** | 4.1e-05 | (A_f + A_t) / M^0.67  (Christiansen 1998) |
| **Est. Body Length** | 9.09 m | Allometric from femur length |
| **Est. Neck Length** | 1.0 m | Allometric from femur length |
| **Est. Tail Length** | 2.54 m | Allometric from femur length |

## 6 Complete Bone Manifest

| Element | L (mm) | D (mm) | Vol cm³ | Mass g | Method | Conf | SF |
|---|---|---|---|---|---|---|---|
| coracoid | 235.8 | 172.5 | 2811.12 | 3509.1 | Constraint | 0.40 | 15.00 |
| femur | 1010.0 | 125.0 | 6321.23 | 7890.8 | Measured | 0.88 | 10.52 |
| femur_right | 1010.0 | 125.0 | 6321.23 | 7890.8 | Symmetry | 0.84 | 10.52 |
| fibula | 860.0 | 40.0 | 389.05 | 408.0 | Measured | 0.88 | 15.00 |
| fibula_right | 860.0 | 40.0 | 389.05 | 408.0 | Symmetry | 0.84 | 15.00 |
| humerus | 620.0 | 88.0 | 1923.17 | 2400.7 | Measured | 0.88 | 5.91 |
| humerus_right | 620.0 | 88.0 | 1923.17 | 2400.7 | Symmetry | 0.84 | 5.91 |
| ilium | 780.0 | 148.0 | 6843.50 | 8542.7 | Measured | 0.88 | 15.00 |
| ilium_right | 780.0 | 148.0 | 6843.50 | 8542.7 | Symmetry | 0.84 | 15.00 |
| ischium | 540.0 | 88.0 | 1675.02 | 2090.9 | Measured | 0.88 | 15.00 |
| metacarpal | 168.8 | 42.0 | 44.40 | 36.5 | Constraint | 0.65 | 2.75 |
| metatarsal | 415.0 | 58.0 | 559.20 | 698.0 | Measured | 0.88 | 6.58 |
| phalanx | 106.1 | 34.5 | 15.24 | 11.8 | Constraint | 0.40 | 15.00 |
| pubis | 455.0 | 72.0 | 944.79 | 1179.4 | Measured | 0.88 | 15.00 |
| radius | 498.0 | 55.0 | 603.41 | 753.2 | Measured | 0.88 | 5.24 |
| radius_right | 498.0 | 55.0 | 603.41 | 753.2 | Symmetry | 0.84 | 5.24 |
| rib | 720.0 | 35.0 | 353.29 | 441.0 | Measured | 0.88 | 15.00 |
| scapula | 640.0 | 235.0 | 14157.15 | 17672.4 | Measured | 0.88 | 15.00 |
| scapula_right | 640.0 | 235.0 | 14157.15 | 17672.4 | Symmetry | 0.84 | 15.00 |
| skull | 850.0 | 380.0 | 49163.88 | 61371.3 | Measured | 0.88 | 15.00 |
| tibia | 890.0 | 95.0 | 3217.34 | 4016.2 | Measured | 0.88 | 10.23 |
| tibia_right | 890.0 | 95.0 | 3217.34 | 4016.2 | Symmetry | 0.84 | 10.23 |
| ulna | 520.0 | 58.0 | 700.68 | 874.7 | Measured | 0.88 | 5.88 |
| ulna_right | 520.0 | 58.0 | 700.68 | 874.7 | Symmetry | 0.84 | 5.88 |
| vertebra | 195.0 | 128.0 | 1279.72 | 1597.5 | Measured | 0.88 | 15.00 |